# 2020-02-19 需求确认

### 业务流程

- 创建合约：
  1. 创建带初始化参数的合约，部署合约 (address[] _owners, uint _required)；
  2. 调用 JUB_SignContractETH 执行创建合约，其中，to 为nil，input 为步骤 1 的结果。
- 合约调用：
  - submitTransaction：
    - ETH 交易
      1. 创建自定义合约交易的 abi，使用 JUB_BuildContractWithAddrAmtData；
      2. 调用 JUB_SignContractETH 执行合约调用，其中，to 为自定义合约地址，input 为步骤1 的结果。
    - 代币 交易
      1. 创建自定义合约代币交易的 abi，使用接口 JUB_BuildERC20AbiETH；
      2. 创建合约调用 abi，使用接口 JUB_BuildContractWithAddrAmtData，其中，data 为步骤1 的结果；
      3. 调用 JUB_SignContractETH 执行合约调用，其中，to 为自定义合约地址，input 为步骤2 的结果。
  - confirmTransaction/revokeTransaction：
    - 创建合约调用 abi，使用接口 **JUB_BuildContractWithTxID**；
    - 调用 JUB_SignContractETH 执行合约调用，其中，to 为自定义合约地址，input 为步骤2 的结果。

### 接口修改调整方案

| | 修改前 | 修改后 | 屏显 | 二次确认? |
|---|---|---|---|---|
| ETH 合约交易 | `JUB_RV JUB_SignContractETH`<br>`IN JUB_UINT16 contextID,`<br>`IN BIP32_Path path,`<br>`IN JUB_UINT32 nonce,`<br>`IN JUB_UINT32 gasLimit,`<br>`IN JUB_CHAR_PTR gasPriceInWei,`<br>`IN JUB_CHAR_PTR to,`<br>`IN JUB_CHAR_PTR valueInWei,`<br>`IN JUB_CHAR_PTR input,`<br>`OUT JUB_CHAR_PTR_PTR raw);` | - 创建合约：<br>- to = nil<br>- input =<br>`0x6060...0029`<br>`0000000000000000000000000000000000000000000000000000000000000040 - methodId`<br>`0000000000000000000000000000000000000000000000000000000000000002 - m=2`<br>`0000000000000000000000000000000000000000000000000000000000000003 - n=3`<br>`000000000000000000000000d5e27985e3e7c008af184b9b78ddd9b75d32698c - 用户1`<br>`0000000000000000000000005738c2fb05064ee7aa92a51835c9742a13c1d628 - 用户2`<br>`00000000000000000000000051fe84227e91a5828bffda79c5aa13967da4f8d0 - 用户3`<br><br>- 执行合约调用：<br>- to = 合约地址<br>- input = JUB_BuildContractWithxxx | gas(gasLimit):<br>xxxxxxxx<br>hash(input hash 前 8 个):<br>0xXXXXXXXXXXXXXXXX<br><br>根据具体合约类型做调整 | 否 |
| 合约组织相关接口 | | | | |
| 多签合约相关 | `JUB_BuildContractWithAddr(`<br>`IN JUB_UINT16 contextID,`<br>`IN JUB_CHAR_PTR methodID,`<br>`IN JUB_CHAR_PTR address`<br>`);`<br>`methodID + address`<br>`addManager:`<br>`0x2d06177a000000000000000000000000\|`<br>`fe3d05a492623b397483c79409643bbfe07155cf`<br>`removeManager:`<br>`0xac18de43000000000000000000000000\|`<br>`fe3d05a492623b397483c79409643bbfe07155cf` | ~~JUB_BuildContractWithAddr(~~<br>~~IN JUB_UINT16 contextID,~~<br>~~IN JUB_CHAR_PTR methodID,~~<br>~~IN JUB_CHAR_PTR address~~<br>~~);~~<br>~~methodID + address~~ | - | - |
| 多签合约相关 | `JUB_BuildContractWithAddrAmt(`<br>`IN JUB_UINT16 contextID,`<br>`IN JUB_CHAR_PTR methodID,`<br>`IN JUB_CHAR_PTR address,`<br>`IN JUB_CHAR_PTR amount`<br>`);`<br>`methodID + address + amount`<br>`TransferTo:`<br>`0x2ccb1b30000000000000000000000000\|`<br>`404c60ea5661eac15c10f4aafcf0b57dfc116d67\|`<br>`0000000000000000000000000000000000000000`<br>`000000000000000000000001` | `JUB_BuildContractWithAddrAmtData(`<br>`IN JUB_UINT16 contextID,`<br>`IN JUB_CHAR_PTR methodID,`<br>`IN JUB_CHAR_PTR address,`<br>`IN JUB_CHAR_PTR amount,`<br>`IN JUB_CHAR_PTR data`<br>`);`<br>`methodID + address + amount + data`<br>`submitTransaction(主币交易):`<br>`0xc6427474\|`<br>`000000000000000000000000bcebf68ac0dda14f893b21b293bd867fabd910d6\| to 地址 (32)`<br>`00000000000000000000000000000000000000000000000000000000000003e8\| amount(32)`<br>`0000000000000000000000000000000000000000000000000000000000000060\| 0x60(32)`<br>`0000000000000000000000000000000000000000000000000000000000000001\| len=1(32)`<br>`0000000000000000000000000000000000000000000000000000000000000000 data=0x00(32)`<br><br>`submitTransaction(代币交易):`<br>`0xc6427474\|`<br>`000000000000000000000000412e48fefbb13345a9905795e48e13461af1c4ee\| 代币合约地址 (32)`<br>`0000000000000000000000000000000000000000000000000000000000000000\| ETH value = 0(32)`<br>`0000000000000000000000000000000000000000000000000000000000000060\| 0x60(32)`<br>`0000000000000000000000000000000000000000000000000000000000000044\| len=0x44(32)`<br>`a9059cbb\| erc-20 methodID`<br>`000000000000000000000000a85a8e32fdb79f411715110655d81659055c3832\| 代币 to 地址 (32)`<br>`000000000000000000000000000000000000000000000000000003220e14900\| 代币 amount(32)`<br>`00000000000000000000000000000000000000000000000000000000 长度为 28 字节的0` | data 域为主币ETH或代币交易信息，是敏感信息，建议显示方式同 ETH/ETH token。<br><br>Confirmsending<br>xxx ETH<br>Maximum fee<br>xxx ETH<br>To address:<br>xxx<br><br>Confirmsending<br>xxx mfcc<br>Maximum fee<br>xxx ETH<br>To address:<br>xxx | 是 |
| 多签合约相关 | `JUB_BuildContractWithTxID(`<br>`IN JUB_UINT16 contextID,`<br>`IN JUB_CHAR_PTR methodID,`<br>`IN JUB_CHAR_PTR`<br>`transactionID`<br>`);`<br>`methodID + transactionID`<br>`signTransaction:`<br>`0x5d9ec210\|`<br>`050cdc001543a2e7c74ba06b345285e9c71fb0b4c`<br>`d94d410287624dac51655f6`<br>`deleteTransaction:`<br>`0xfcf45c0f\|`<br>`050cdc001543a2e7c74ba06b345285e9c71fb0b4c`<br>`d94d410287624dac51655f6` | `JUB_BuildContractWithTxID(`<br>`IN JUB_UINT16 contextID,`<br>`IN JUB_CHAR_PTR methodID,`<br>`IN JUB_CHAR_PTR transactionID`<br>`);`<br>methodID + transactionID<br>**用于执行 confirmTransaction 和 revokeConfirmation，因为两者入参是一样的，用不同的 methodID 即可。**<br>`confirmTransaction:`<br>`0xc01a8c84\|050cdc001543a2e7c74ba06b345285e9c71fb0b4cd94d410287624dac51655f6`<br>`executeTransaction:`<br>`0xee22610b\|050cdc001543a2e7c74ba06b345285e9c71fb0b4cd94d410287624dac51655f6`<br>`revokeConfirmation:`<br>`0x20ea8d86\|050cdc001543a2e7c74ba06b345285e9c71fb0b4cd94d410287624dac51655f6` | gas(gasLimit):<br>xxxxxxxx<br>address:<br>0xXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX | 否 |
| 多签合约相关 | `JUB_BuildContractWithAmt(`<br>`IN JUB_UINT16 contextID,`<br>`IN JUB_CHAR_PTR methodID,`<br>`IN JUB_CHAR_PTR amount`<br>`);`<br>`methodID + amount`<br>`withdraw:`<br>`0x2e1a7d4d\|`<br>`0000000000000000000000000000000000000000`<br>`000000000000000000000001` | ~~JUB_BuildContractWithAmt(~~<br>~~IN JUB_UINT16 contextID,~~<br>~~IN JUB_CHAR_PTR methodID,~~<br>~~IN JUB_CHAR_PTR amount~~<br>~~);~~<br>~~methodID + amount~~ | - | - |
| | 修改前 | 修改后 | 屏显 | 二次确认? |

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有以下几种 input：
- Add/Remove Manager（参数：address）
- TransferTo（参数： address to + amount）
- Sign/Remove Transaction（参数：transactionID）
- Withdraw（参数：amount）

这 6 个 input，抽象出 4 种 input 格式，并提供拼凑这 4 中 input 的 SDK 接口，分别是：
- methodID + address -> 组 Add/Remove Manager
- methodID + address + amount -> 组 TransferTo
- methodID + transactionID -> 组 Sign/Remove Transaction
- methodID + amount -> 组 Withdraw

然后，调用现有 JUB_SignContractETH () 接口的 input 入参进行显示和签名。

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关于显示，需要确认以下几点：

| | | to | 屏幕显示格式： | 二次确认? |
|---|---|---|---|---|
| ETH 交易 | `JUB_RV JUB_SignTransactionETH`<br>`IN JUB_UINT16 contextID,`<br>`IN BIP32_Path path,`<br>`IN JUB_UINT32 nonce,`<br>`IN JUB_UINT32 gasLimit,`<br>`IN JUB_CHAR_PTR gasPriceInWei,`<br>`IN JUB_CHAR_PTR to,`<br>`IN JUB_CHAR_PTR valueInWei,`<br>`IN JUB_CHAR_PTR input,`<br>`OUT JUB_CHAR_PTR_PTR raw);` | 目的地址 | Confirm sending<br>xxx ETH<br>Maximum fee<br>xxx ETH<br>To address:<br>0xXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX | 是 |
| ETH token 交易 | `JUB_RV JUB_BuildERC20AbiETH`<br>`IN JUB_UINT16 contextID,`<br>`IN JUB_CHAR_PTR tokenName,`<br>`IN JUB_UINT16 unitDP,`<br>`IN JUB_CHAR_PTR contractAddress,`<br>`IN JUB_CHAR_PTR tokenTo,`<br>`IN JUB_CHAR_PTR tokenValue,`<br>`OUT JUB_CHAR_PTR_PTR abi);`<br>拼 input | token 合约地址 | Confirm sending<br>xxx RVT<br>Maximum fee<br>xxx ETH<br>To address:<br>0xXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX | 是 |
| ETH 合约交易 | `JUB_RV JUB_SignContractETH`<br>`IN JUB_UINT16 contextID,`<br>`IN BIP32_Path path,`<br>`IN JUB_UINT32 nonce,`<br>`IN JUB_UINT32 gasLimit,`<br>`IN JUB_CHAR_PTR gasPriceInWei,`<br>`IN JUB_CHAR_PTR to,`<br>`IN JUB_CHAR_PTR valueInWei,`<br>`IN JUB_CHAR_PTR input,`<br>`OUT JUB_CHAR_PTR_PTR raw);` | 创建合约时=nil；<br>调用合约时=合约地址 | | 否 |
| 多签合约相关 | `0x6080...0029` | nil | gas(gasLimit):<br>xxxxxxxx<br>hash(input hash 前 8 个):<br>0xXXXXXXXXXXXXXXXX | 否 |
| 多签合约相关 | `JUB_BuildContractWithAddr(`<br>`IN JUB_UINT16 contextID,`<br>`IN JUB_CHAR_PTR methodID,`<br>`IN JUB_CHAR_PTR address`<br>`);`<br>methodID + address<br>abi:<br>`addManager:`<br>`0x2d06177a000000000000000000000000\|fe3d05a492623b397483c79409643bbfe07155cf`<br>`removeManager:`<br>`0xac18de43000000000000000000000000\|fe3d05a492623b397483c79409643bbfe07155cf` | 多签合约地址 | gas(gasLimit):<br>xxxxxxxx<br>address:<br>0xXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX | 否 |
| 多签合约相关 | `JUB_BuildContractWithAddrAmt(`<br>`IN JUB_UINT16 contextID,`<br>`IN JUB_CHAR_PTR methodID,`<br>`IN JUB_CHAR_PTR address,`<br>`IN JUB_CHAR_PTR amount`<br>`);`<br>methodID + address + amount<br>abi:<br>`TransferTo:`<br>`0x2ccb1b30000000000000000000000000\|404c60ea5661eac15c10f4aafcf0b57dfc116d67\|0000000000000000000000000000000000000000000000000000000000000001` | 多签合约地址 | gas(gasLimit):<br>xxxxxxxx<br>address:<br>0xXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX<br>amount:<br>xxx ETH | 否 |
| 多签合约相关 | `JUB_BuildContractWithTxID(`<br>`IN JUB_UINT16 contextID,`<br>`IN JUB_CHAR_PTR methodID,`<br>`IN JUB_CHAR_PTR transactionID`<br>`);`<br>`methodID + transactionID`<br>`signTransaction:`<br>`0x5d9ec210\|`<br>`050cdc001543a2e7c74ba06b345285e9c71fb0b4cd94d410287624dac51655f6`<br>`deleteTransaction:`<br>`0xfcf45c0f\|`<br>`050cdc001543a2e7c74ba06b345285e9c71fb0b4cd94d410287624dac51655f6` | 多签合约地址 | gas(gasLimit):<br>xxxxxxxx<br>trxID（32字节字符串）：<br>0xXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX | 是 |
| 多签合约相关 | `JUB_BuildContractWithAmt(`<br>`IN JUB_UINT16 contextID,`<br>`IN JUB_CHAR_PTR methodID,`<br>`IN JUB_CHAR_PTR amount`<br>`);`<br>methodID + amount<br>`withdraw:`<br>`0x2e1a7d4d\|`<br>`0000000000000000000000000000000000000000000000000000000000000001` | 多签合约地址 | gas(gasLimit):<br>xxxxxxxx<br>amount:<br>xxx ETH | 否 |
| | | to | 屏幕显示格式： | 二次确认? |

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```
[{"constant":false,"inputs":[{"name":"to","type":"address"},{"name":"amount","type":"uint256"}],"name":"transferTo","outputs":
[],"payable":false,"stateMutability":"nonpayable","type":"function"},{"constant":false,"inputs":
[{"name":"manager","type":"address"}],"name":"addManager","outputs":
[],"payable":false,"stateMutability":"nonpayable","type":"function"},{"constant":false,"inputs":
[{"name":"amount","type":"uint256"}],"name":"withdraw","outputs":
[],"payable":false,"stateMutability":"nonpayable","type":"function"},{"constant":false,"inputs":
[{"name":"transactionId","type":"uint256"}],"name":"signTransaction","outputs":
[],"payable":false,"stateMutability":"nonpayable","type":"function"},{"constant":false,"inputs":
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[{"name":"","type":"uint256[]"}],"payable":false,"stateMutability":"view","type":"function"},{"constant":false,"inputs":
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{"indexed":false,"name":"amount","type":"uint256"}],"name":"DepositFunds","type":"event"},{"anonymous":false,"inputs":
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{"indexed":false,"name":"amount","type":"uint256"}],"name":"TransferFunds","type":"event"},{"anonymous":false,"inputs":
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{"indexed":false,"name":"amount","type":"uint256"},
{"indexed":false,"name":"transactionId","type":"uint256"}],"name":"TransactionCreated","type":"event"}]
```

```
0x608060405234801561001057600080fd5b50336000806101000a81548173ffffffffffffffffffffffffffffffffffffffff021916908373fffffffff
ffffffffffffffffffffffffffffff1602179055506110c3806100606000396000f30060806040526004361061008e576000357c010000000000000000
00000000000000000000000000000000000000000900463ffffffff1680632ccb1b30146100fb5780632d06177a146101485780632e1a7d4d1461018b578
0635d9ec210146101b8578063ac18de43146101e5578063cd0a314b14610228578063d11db83f14610253578063fcf45c0f146102bf575b7fd241e73300
212f6df233a8e6d3146b88a9d4964e06621d54b5ff6afeba7b1b8833346040518083 73ffffffffffffffffffffffffffffffffffffffff1673fffffffff
ffffffffffffffffffffffffffffff16815260200182815260200192505050604051809103 90a1005b34801561010757600080fd5b5061014660048036
0381019080803573ffffffffffffffffffffffffffffffffffffffff169060200190929190803590602001909291905050506102ec565b005b348015610
15457600080fd5b50610189600480360381019080803573ffffffffffffffffffffffffffffffffffffffff169060200190929190505050610619565b00
5b34801561019757600080fd5b506101b66004803603810190808035906020019092919050505061 06cf565b005b3480156101c457600080fd5b506101e
3600480360381019080803590602001909291905050506107 90565b005b3480156101f157600080fd5b50610226600480360381019080803573ffffffff
ffffffffffffffffffffffffffffffff16906020019092919050505061 0b4c565b005b34801561023457600080fd5b5061023d610c03565b60405180828
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35b3480156102cb57600080fd5b506102ea60048036038101908080359060200190929190505050610de2565b005b60006102f6610fee565b6000809054
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ffffffffffffffffffffffffff16908173ffffffffffffffffffffffffffffffffffffffff168152505083816020019073fffffffffffffffffffffffff
fffffffffffffff16908173ffffffffffffffffffffffffffffffffffffffff168152505082816040018181525050600081606001906 0ff16908160ff16
81525050806003600084815260200190815260200160002060008201518160000160006101000a81548173fffffffffffffffffffffffffffffffffffff
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ffffffffffffffffffffff021916908373ffffffffffffffffffffffffffffffffffffffff160217905550604082015181600201556060820151816003 0
160006101000a81548160ff021916908360ff160217905550905050600482908060018154018082558091505090600182039060005260206000200160 00
909192909190915055507f41d8c89c4a25824354aa1f198d80d06f2f93f903a2d180d1ea20429a4678c4a4338585856040518085 73fffffffffffffffff
fffffffffffffffffffffff1673ffffffffffffffffffffffffffffffffffffffff1681526020018473ffffffffffffffffffffffffffffffffffffffff
1673ffffffffffffffffffffffffffffffffffffffff16815260200183815260200182815260200194505050505060405180910390a150505050565b337
3ffffffffffffffffffffffffffffffffffffffff1660008090549061 01000a900473ffffffffffffffffffffffffffffffffffffffff1673ffffffffff
ffffffffffffffffffffffffffffff16141515610674576 00080fd5b6001806000 8373ffffffffffffffffffffffffffffffffffffffff1673fffffffff
fffffffffffffffffffffffffffffff16815260200190815260200160002060006101000a81548160ff021916908360ff16021790555050565b60008090
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fffffffffffffffffffffffff161480610778575060018060003373ffffffffffffffffffffffffffffffffffffffff1673ffffffffffffffffffffffff
ffffffffffffffff16815260200190815260200160002060009054906101000a900460ff1660ff16145b15156107835 7600080fd5b61078d33826102ec5
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163373ffffffffffffffffffffffffffffffffffffffff16148061083b575060018060003373ffffffffffffffffffffffffffffffffffffffff1673fff
ffffffffffffffffffffffffffffffffffffff16815260200190815260200160002060009054906101000a900460ff1660ff16145b15156108465 7600080
fd5b600360008381526020019081526020016000209050806000016000905490610 1000a900473ffffffffffffffffffffffffffffffffffffffff1673f
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52602001908152602001600020600090549061 01000a900460ff1660ff16141515156109655 7600080fd5b60018160040160003373fffffffffffffffff
fffffffffffffffffffffff1673ffffffffffffffffffffffffffffffffffffffff16815260200190815260200160002060006101000a81548160ff0219
16908360ff16021790555080600301600081819054906101000a900460ff16809291906001019190610 1000a81548160ff021916908360ff16021790555
05060038160030160009054906101000a900460ff1660ff16101515610b485780600201543073ffffffffffffffffffffffffffffffffffffffff163110
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1681526020018281526020019250505060405180910390a1610b4782610de2565b5b5050565b3373ffffffffffffffffffffffffffffffffffffffff166
00080905490610 1000a900473ffffffffffffffffffffffffffffffffffffffff1673ffffffffffffffffffffffffffffffffffffffff16141515610ba7
57600080fd5b6000600160008373ffffffffffffffffffffffffffffffffffffffff1673ffffffffffffffffffffffffffffffffffffffff16815260200
190815260200160002060006101000a81548160ff021916908360ff16021790555050565b6000806000905490610 1000a900473ffffffffffffffffffff
ffffffffffffffffffff1673ffffffffffffffffffffffffffffffffffffffff163373ffffffffffffffffffffffffffffffffffffffff161480610cae5
75060018060003373ffffffffffffffffffffffffffffffffffffffff1673ffffffffffffffffffffffffffffffffffffffff1681526020019081526020
0160002060009054906101000a900460ff1660ff16145b1515610cb957600080fd5b3073ffffffffffffffffffffffffffffffffffffffff16319050905
65b60606000809054906101000a900473ffffffffffffffffffffffffffffffffffffffff1673ffffffffffffffffffffffffffffffffffffffff163373
ffffffffffffffffffffffffffffffffffffffff161480610d8157506001806000 3373ffffffffffffffffffffffffffffffffffffffff1673fffffffff
fffffffffffffffffffffffffffffff16815260200190815260200160002060009054906101000a900460ff1660ff16145b1515610d8c57600080fd5b60
0480548060200260200160405190810160405280929190818152602001828054801561 0dd857602002820191906000526020600020905b8154815260200
19060010190808311610dc4575b505050505090509056 5b6000806000809054906101000a900473ffffffffffffffffffffffffffffffffffffffff1673
ffffffffffffffffffffffffffffffffffffffff163373ffffffffffffffffffffffffffffffffffffffff161480610e8e575060018060003373fffffff
fffffffffffffffffffffffffffffffff1673ffffffffffffffffffffffffffffffffffffffff16815260200190815260200160002060009054906101 00
0a900460ff1660ff16145b1515610e9957600080fd5b60009150600090505b600480549050811015610f2e578160ff1660011415610efa5760048181548
1101515610eca57fe5b9060005260206000200154600460018303815481101515610ee757fe5b9060005260206000200181905550610f21565b60048181
5481101515610f0957fe5b906000526020600020015483141561 0f2057600191505b5b8080600101915050610ea2565b60046001600480549050038154 8
1101515610f4557fe5b90600052602060002001600090556004805480919060019003610f689190611046565b5060036000848152602001908152602001
6000206000808201600061 01000a81549073ffffffffffffffffffffffffffffffffffffffff0219169055600182016000610 1000a81549073fffffffff
fffffffffffffffffffffffffffffff0219169055600282016000905560038201600061 01000a81549060ff02191690555050505050565b608060405190
81016040528060007 3ffffffffffffffffffffffffffffffffffffffff168152602001600073ffffffffffffffffffffffffffffffffffffffff1681526
0200160008152602001600060ff1681525090565b81548183558181111561106d5781836000526020600020918201910161106c9190611072565b5b5050
50565b61109491905b808211156110905760008160009055506001016110 78565b5090565b905600a165627a7a72305820151bfad8db08d2977be2d0bff
ef557bdbd15f37f5b595566580a3b1ebfa3d58f0029
```

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**abi:**

**0xa9059cbb000000000000000000000000ef31dec147dcdcd64f6a0abfa7d441b62a216bc9000000000000000000000000000000000000000000000000112210f47de98115**

**tx:**

**0xf8640c8502540be400830249f0943d1ba9be9f66b8ee101911bc36d3fb562eac22448026a03d118b67df7e5b267606e09342d80d374ad09e0357bdc7abb80a26af36421cd2a0607063c634875ec1528d901043e682**
**b9a7d8cc4262c6c674f295be66a52f8f26**

**f8640c8502540be400830249f094 3d1ba9be9f66b8ee101911bc36d3fb562eac2244 8026a0**
**3d118b67df**
**7e5b267606e09342d80d**
**374ad09e0357bdc7abb8**
**0a26af36421cd2a06070**
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**43e682b9a7d8cc4262c6**
**c674f295be66a52f8f26**